

6 Historical Counting Systems

6.2 Positional Systems

As we have noticed, the notion of ‘position’ is extremely valuable, but it took a long time for this idea to develop. Let’s discuss some of the basics of what a positional system is.

Write a number, explain briefly what each position is, this is pretty basic. Note that ancient civilizations did not necessarily use 10 as their base.

If we choose 8 as our base for a positional system, what are the values of each position?

$8^0 = 1, 8^1 = 8, 8^2 = 64, 8^3 = 512, 8^4 = 4096, \dots$

Write the number 502 (which is in base 8) in base 10.

make a notational note that we will always write non-base 10 numbers with the base as a subscript. So this number should be written 502_8 .

Write the number 3324_5 in base 10.

Write the number 10101_2 in base 10.